

VPS72 p.Lys115Arg | JLF V3 | 27 aa

disclosed minimal epitope ERALLPLEL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

MHC-I: 5 shown / 12 eligible (top-10 cap)

C01:02 0.31%

C07:01 0.25%

B27:05 1.1%

C07:01 1.7%

B08:01 0.89%

MHC-II: 5 shown / 10 eligible (top-10 cap)

DR03:01 3.1%

DR03:01 2.7%

DR03:01 2.2%

DR03:01 3.1%

DR03:01 2.3%

ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure

G S S Q K A R E E R A L L P L E L Q D D G S D S R K S

Intact SLP bond 2 - free N terminus

DPP4 / qPISA
N-terminal 2-residue removal

+0.343 log2 depletion

MME / neprilysin
Internal hydrophobic preference

4 motif matches / 26 assessed bonds - partial recognition rule

FAP - internal
Gly-Pro|non-Pro recognition

0 motif matches / 26 assessed bonds - partial recognition rule

FAP - N-terminal
Remove 2 residues: X-Pro|non-Pro

0 motif matches / 1 assessed bond - partial recognition rule

ANPEP / CD13
exposed N-terminus only

0 motif matches / 1 assessed bond - partial recognition rule

ENPEP
exposed N-terminus only

0 motif matches / 1 assessed bond - partial recognition rule

M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.